

Kritika Pandit



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Education

Carleton College

Expected June 2026

Bachelor of Arts in Computer Science & Statistics (GPA: 3.71 / 4.00)

Northfield, Minnesota

- **Relevant Coursework:** Data Structures & Algorithms, Software Design, Systems, Linear Algebra, Discrete Math, Calculus Series, Data Science, Theory of Computing, Mobile App Development, Computer Security, Probability

Experience

Software Engineering Fellow | HeadStarter AI, San Francisco, CA

Jul 24, Sep 24

- Developed an AI-powered Pantry Tracker App using **Google Gemini API**, reducing grocery waste by 20%, and built AI Flashcards with **Stripe** and **Clerk**, scaling to 500+ weekly active users.
- Created an AI ChatBot in **React**, handling 300+ conversations weekly with a 4.8/5 satisfaction rating, and launched Rate My Professor AI, where 85 % of users reported better course decisions.
- Participated in the fellowship **hackathon**, collaborating on an AI-powered productivity tool and gaining hands-on experience in rapid **prototyping** and pitching innovative solutions.

DevSquad Junior Developer | Academic Technology, Carleton College

Apr 24 - Sep 24

- Replaced Carleton's GPT-3.5 ITS helpdesk chatbot with an open-source **Llama3** model, improving support response times by 40% and ensuring data security through **PrivateGPT**.
- Enhanced model performance and efficiency by implementing GPU acceleration, integrating the **pgpt.Python** SDK to create a robust real-time server with streaming capabilities, and enabling **multi-document injection** to support over 1,000 monthly interactions.

ML Research Assistant | Hooper Lab, Carleton College

Jun 24 - Aug 24

- Improved pose tracking and behavior analysis accuracy by 30% through the creation of skeletal models with **SLEAP** deep learning software, advancing Drosophila research outcomes.
- Streamlined data workflows using **TensorFlow**, **NumPy**, and **Matplotlib**, reducing data preprocessing and model training time by 25% for more efficient and accurate analysis.

Junior Software Developer | Virtual Viking Longship Project

Jul 23 - Apr 24

- Developed interactive Viking Age **VR simulations** for enhanced historical education, increasing engagement in classroom demos by 50%.
- Designed a **Unity** based rowing mini-game and implemented **mixed reality** elements, enabling immersive learning experiences that received positive feedback from 95% of users in testing.

Projects

Carls Swap Prototype

Jan 25- Present

- Designing a sustainable campus exchange platform to reduce waste through streamlined item sharing, integrating user-driven features like reservations, secure payments, and Carleton-exclusive access.
- Conducted **axial coding** on 15+ stakeholder transcripts using **NVivo** to derive 20+ design requirements, translating insights into **Figma** wireframes, user flow diagrams, and interactive prototypes.

Travel Journal App

Nov 24- Dec 24

- Designed and developed an interactive travel journal app with expense tracking, itinerary management, and multimedia storage, enhancing user engagement by 70%.
- Built and optimized location-based features using **Google Maps API** and **Jetpack Compose**, improving trip planning and navigation.
- Conducted user testing, iterated on **UI/UX** based on feedback, and implemented **Firebase** and **Room Database** for seamless data storage and retrieval.

Carls Chat Website

Apr 24- Jun 24

- Led a team of 4 to develop, a **Flask-based** platform enabling real-time chat, avatar customization, & location based activities, boosting student engagement by 30%.
- Deployed the application using **Docker** and implemented **WebSocket technology**, supporting 100+ users and achieving 85% user satisfaction.

House Price Prediction Model

Apr 24 - Apr 24

- Created **Shiny app** in **R** to predict California house prices, using machine learning models and exploratory data analysis, including **Random Forest Modeling** and **variable importance analysis**.
- Implemented data preprocessing, spatial joins, and visualization using **ggplot2** and **plotly** to create a user-friendly interface for exploring housing data insights.

Skills

Languages : Python, Java, HTML5/CSS, R, MATLAB, Kotlin, Ruby

Databases : PostgreSQL, SQLite, FireBase, MongoDB

Libraries : React, NumPy, TensorFlow, Pandas, PyTorch, Matplotlib, NLTK

Frameworks : Flask, Bootstrap, Django

Tools : Git, Node.js, Docker, Figma

Languages : Nepali, Hindi